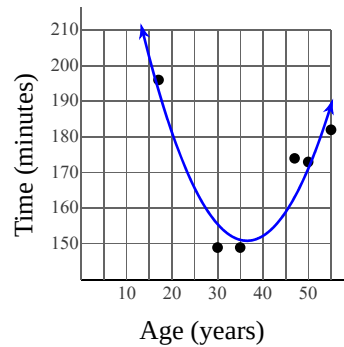


- 3) The time for the fastest runner for their age in the Marine Corps Marathon is given for several ages:

Age	Time (minutes)
17	196
30	149
35	149
47	174
50	173
55	182

This can be modeled by the equation $y = 0.111x^2 - 8.1x + 299$ where x is the age and y is the number of minutes taken.

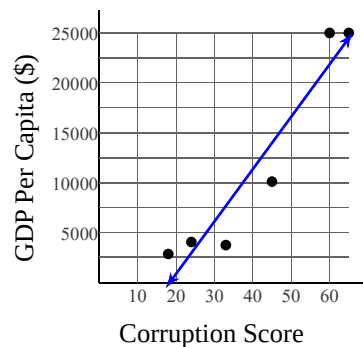


- a) Using this model, what would be the time for the fastest 41-year-old? Round your answer to the nearest hundredth.
- b) According to the model, what would be the time for the fastest 83-year-old? Round your answer to the nearest hundredth.
- c) What age(s) correspond to a time of 171 minutes? Round your answer(s) to the nearest tenth.

- 4) Economists have found that the amount of corruption in a country is correlated to the productivity of that country. Productivity is measured by gross domestic product (GDP) per capita. Corruption is measured on a scale from 0 to 100 with 0 being highly corrupt and 100 being least corrupt:

Corruption Score	GDP Per Capita (\$)
18	2,850
24	4,050
33	3,740
45	10,100
60	25,000
65	25,000

This can be modeled by the equation $y = 525x - 9640$ where x is the corruption score and y is GDP per capita in dollars.



- a) According to the model, what would be the GDP per capita of a country with a corruption score of 38? Round your answer to the nearest dollar.
- b) Using this model, a country with a corruption score of 75 would have what GDP per capita? Round your answer to the nearest dollar.
- c) A GDP per capita of \$17,000 corresponds to what corruption score, according to the model? Round your answer to the nearest whole number.